

Hello Belgium!

Dutch: Hallo

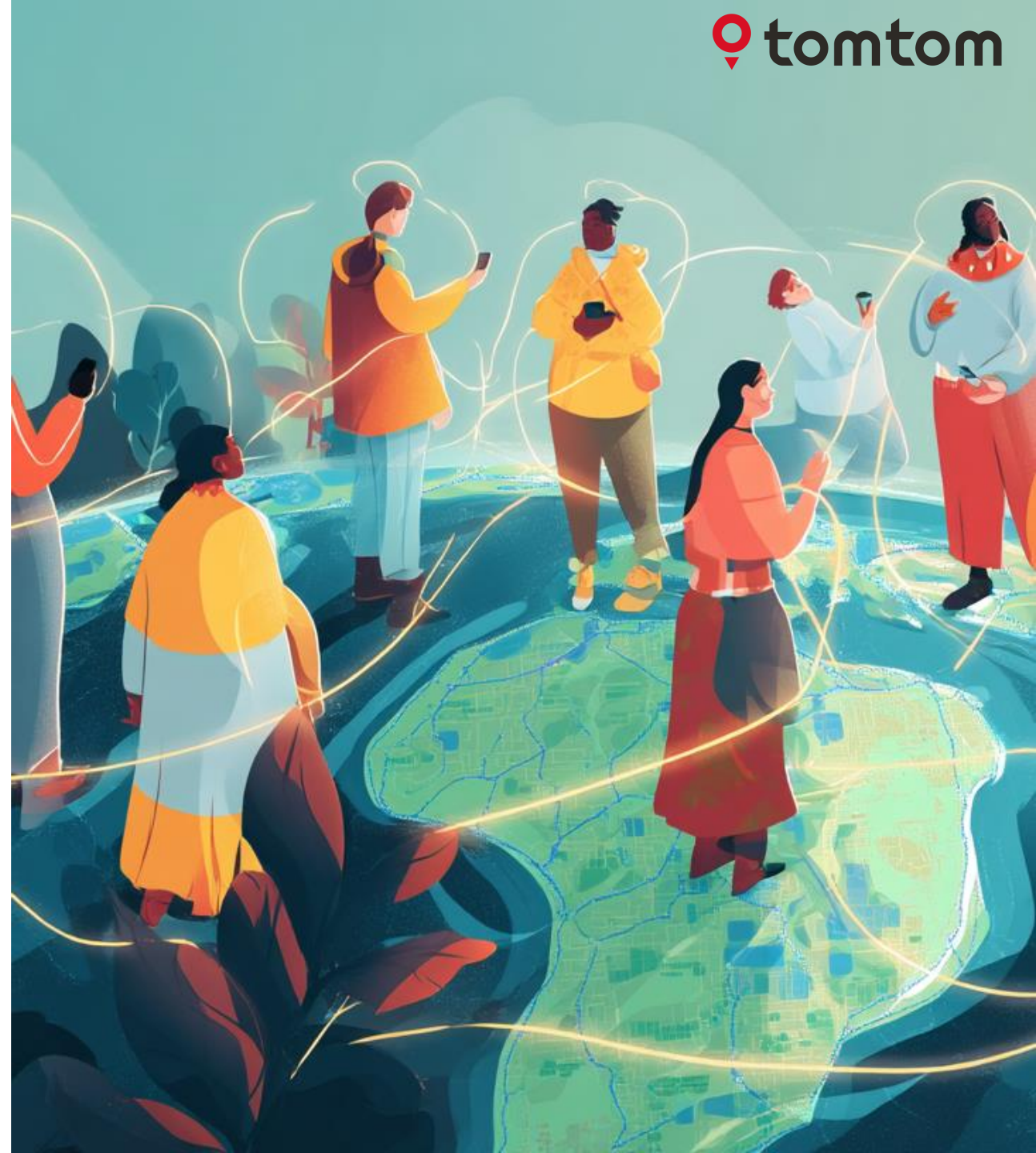
French: Bonjour

German: Hallo



Building Bridges: Corporate-Community Collaboration in Open Source Mapping

Partnerships for shared mapping innovations



Today's Speakers



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TomTom

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The world we are living in, right now

The World We're Mapping Together

Open Empowerment

Built on open maps and shared data sources empowering global communities.

Entities Integration

Diverse entities amplify community impact through partnerships and resources.

Future Together

Focus on sustainable, long-term collaborations driving global positive impact.

Community Collaboration Scale

Over 10 million registered Open Map and Source contributors and developer with millions of daily edits and created coding line showcase vast community engagement.

Impact and Sustainability

Emphasis on maintaining value, reliability, and lasting benefits of collaborative mapping.

Corporate Support Importance

Corporate involvement enhances collaboration and sustainability, supporting the project's ongoing impact.



Why Open Maps & Source Matter

Community-Driven Data Creation

Open maps rely on community-driven data creation, harnessing local knowledge for accurate geographic information.

Global Accessibility

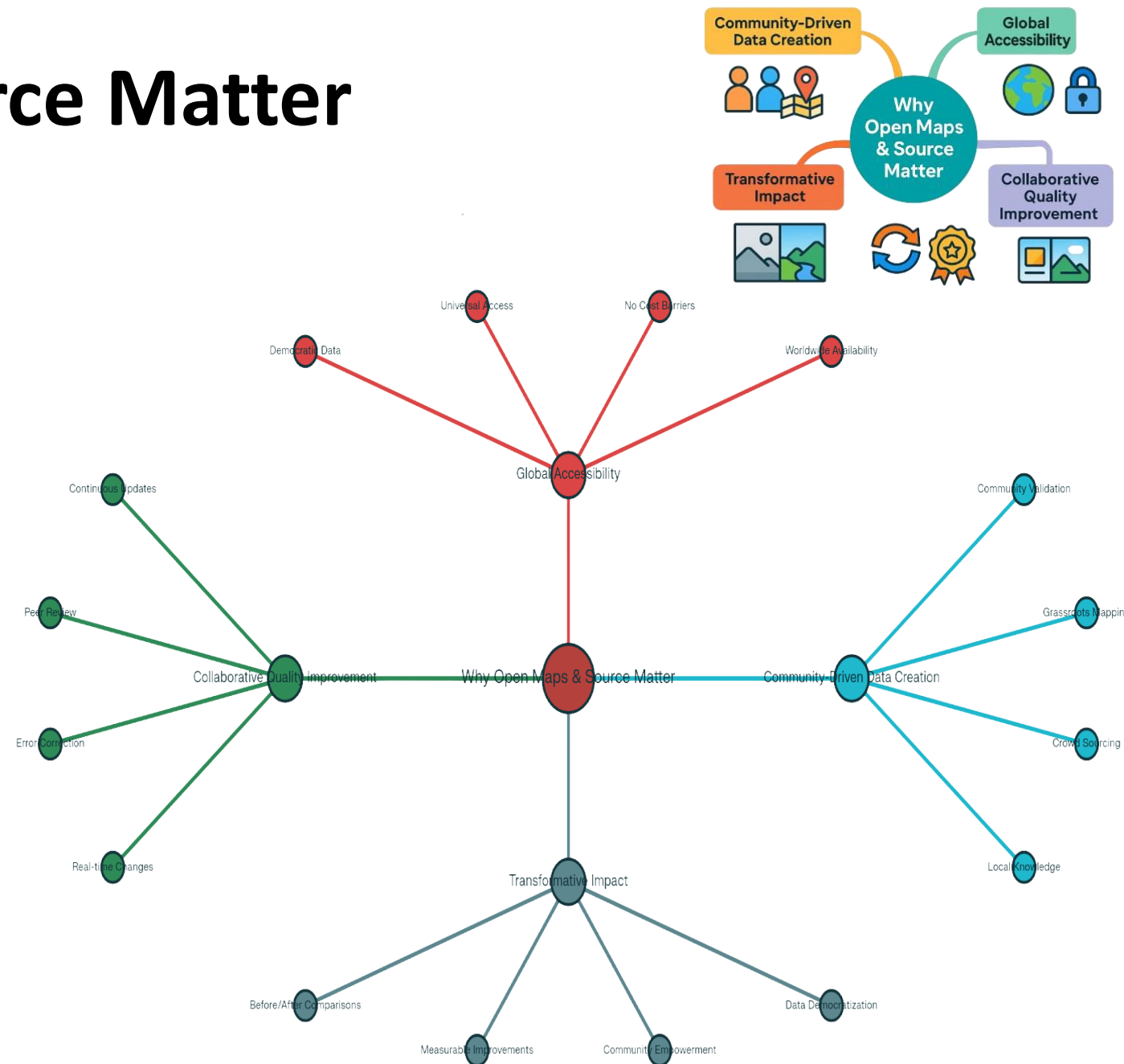
Open maps provide global accessibility, ensuring geographic data is available to everyone worldwide.

Collaborative Quality Improvement

Continuous collaborative efforts improve map quality, making data more reliable and up to date.

Transformative Impact

Visual comparisons before and after community contributions demonstrate the transformative impact of open maps.



The OSGeo Ecosystem

Diverse Project Portfolio

OSGeo hosts essential geospatial tools like QGIS, PostGIS, GeoServer, GDAL/OGR, and MapServer for mapping and analysis.

Community and Initiatives

Initiatives like Geo for All and FOSS4G conferences promote community engagement and open standards development.

Scalable Collaboration Mission

Since 2006, OSGeo provides infrastructure and governance to enable scalable collaborative development worldwide.



OpenStreetMap: The Community Success Story

OpenStreetMap

A Community Success Story spanning over 21 years of growth.

Milestones

Significant expansion with a large global user base and extensive geographic data contributions.

European Community Contributions

Key involvement from countries like Germany, Belgium, Netherlands that strengthen OSM's detail and coverage.

Sustained Engagement:

Ongoing regional leadership and active community participation have driven quality and growth over time.



Observation & Challenges

Findings and Challenges

Economic Value and Policy Impact

Open source projects generate \$8.8 trillion in value but face mixed government policies affecting their growth and governance.

Community Sustainability Crisis

58% of maintainers consider quitting due to burnout, reflecting critical sustainability and support challenges in open source communities.

Contributor Decline and Bias

Open source sees a 20% decline in new contributors, low retention, geographic bias, and an aging maintainer base impacting project vitality.

Technical and Financial Challenges

Legacy dependencies, quality control issues, and funding gaps threaten open source project scalability and long-term viability.



Government & Corporate Impact

Government Policy Impact

Policies like the US OPEN Data Act and EU Data Governance Act boost availability of public datasets and infrastructure support.

Licensing and Privacy Challenges

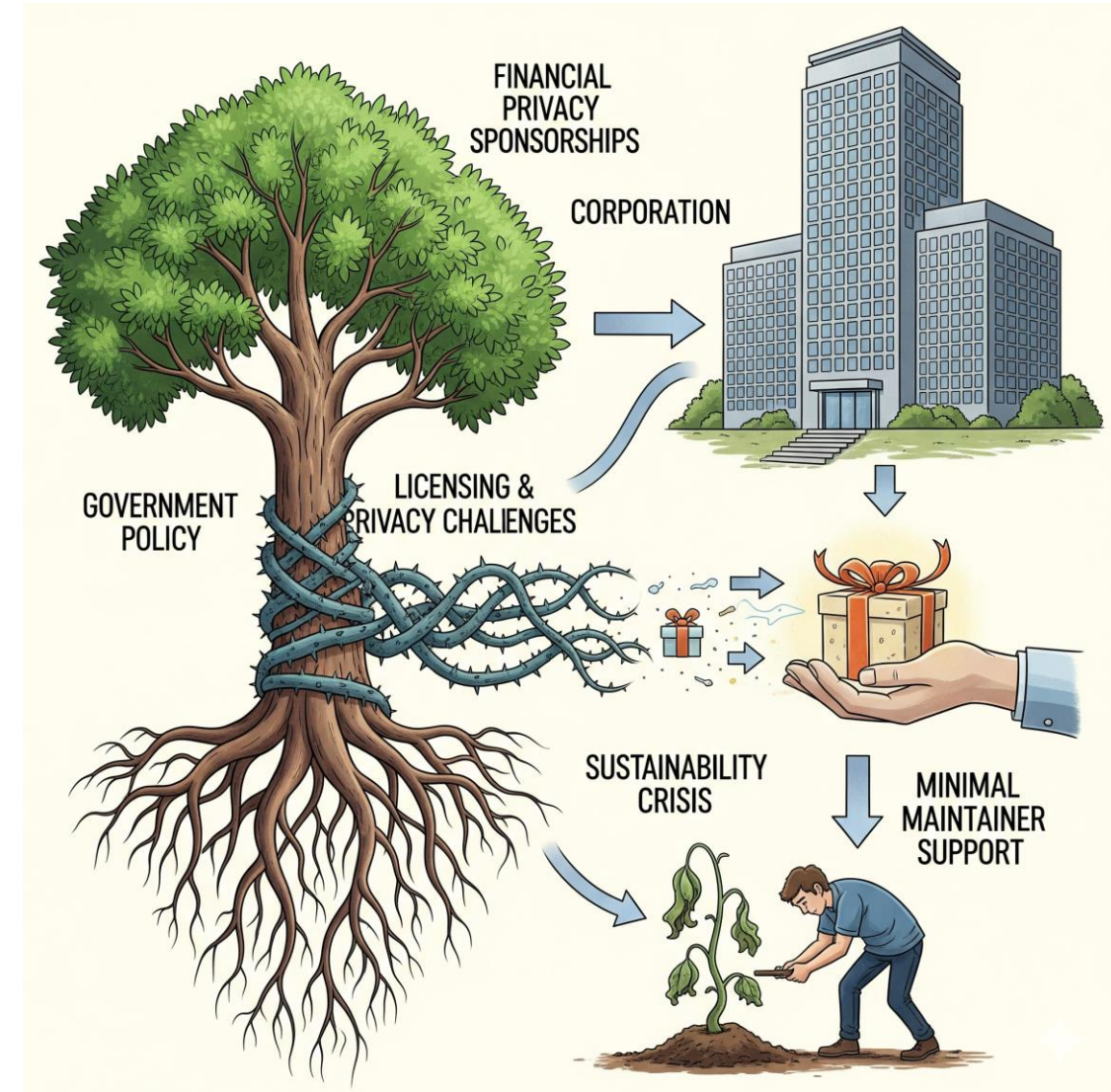
Complex licensing and privacy regulations such as GDPR hinder data integration and impose compliance burdens.

Corporate Contributions and Trends

Corporate contributions peaked in 2020–2021 but have since declined as companies shift to proprietary platforms.

Financial Support and Sustainability

Financial sponsorships have grown, yet sustainability crisis persists with minimal maintainer support.



What can you do

Strategic Interventions for Sustainability

Government Roles

Governments should adopt CC0 licensing, standardize attribution, fund projects, and coordinate internationally for sustainability.

Corporate Engagement

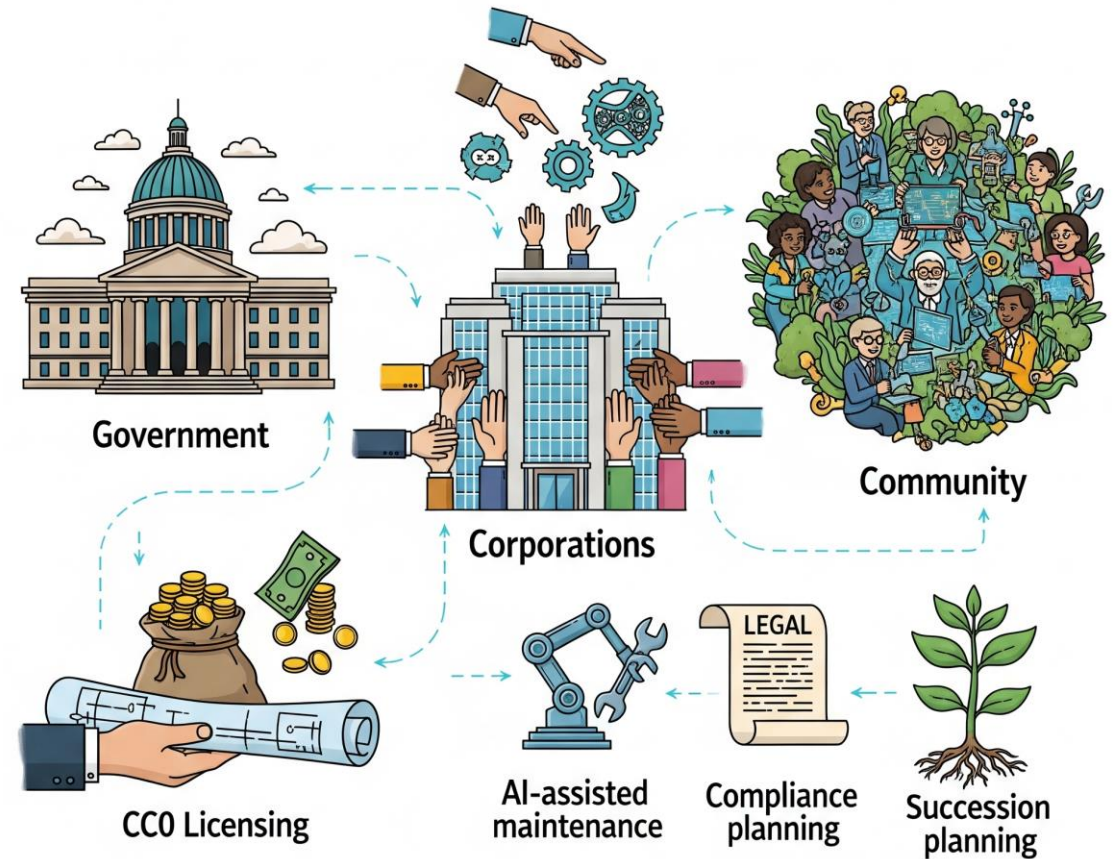
Corporations must commit funding, diversify support, provide infrastructure, mentorship, and allocate employee contribution time.

Community and Ecosystem

Transparency, avoiding extractive practices, diverse engagement, and structured mentorship strengthen community ecosystems.

Sustainability Strategies

AI-assisted maintenance, compliance planning, legal resource pooling, and succession planning ensure long-term project health.



Safe Havens & Sustainability

Technical Safe Havens

Trusted Research Environments and the framework provide secure and controlled development environments for sensitive projects.

Economic Stability

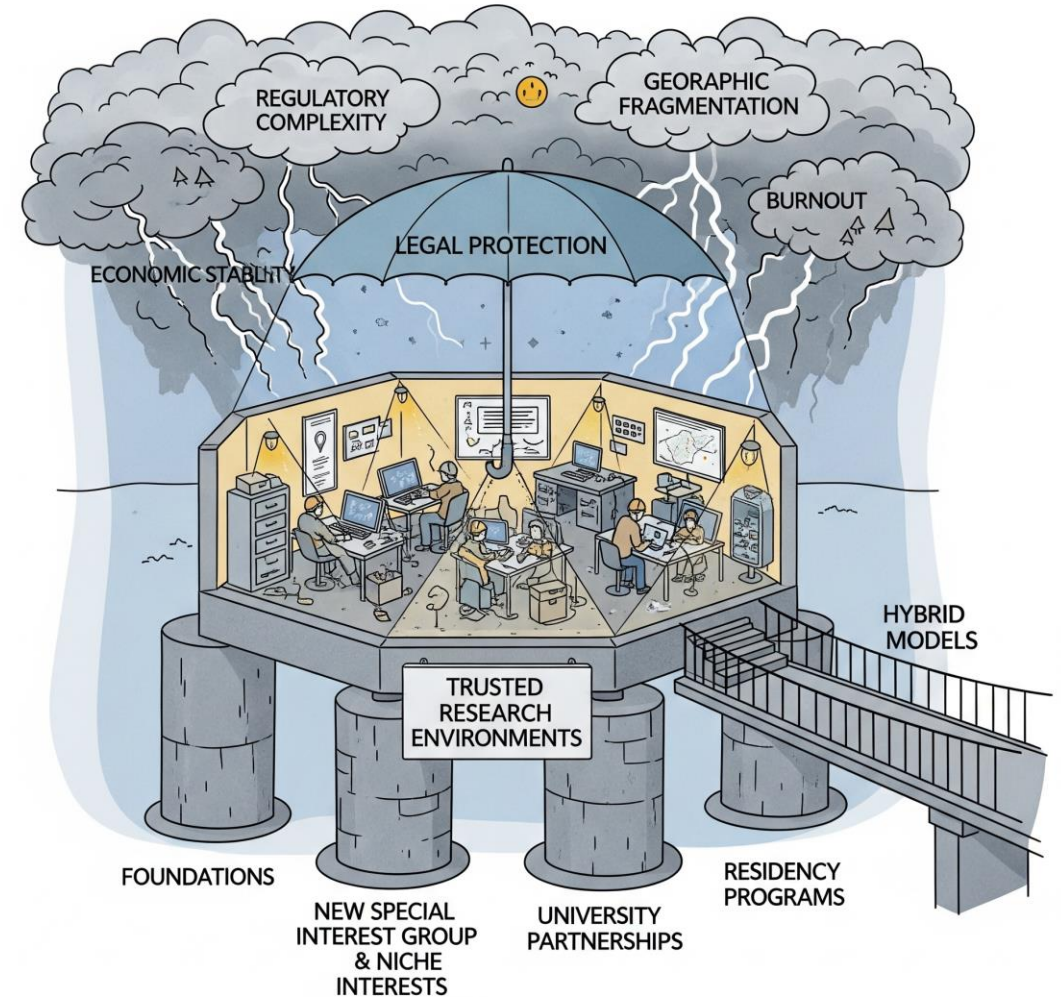
Foundations,, new Special Interest Group, Niche interests, university partnerships, and residency programs provide stable funding and employment for open source developers.

Legal Protection

Legal umbrellas and export control exemptions help developers navigate complex regulations and maintain compliance.

Challenges and Hybrid Models

Addressing regulatory complexity, geographic fragmentation, and burnout through hybrid volunteer-professional collaboration models.



Path Forward for Open Source and Open Maps

Challenges Facing Open Source

Open source projects face maintainer burnout and regulatory complexities but remain viable through strategic adaptation.

Role of Governments and Corporations

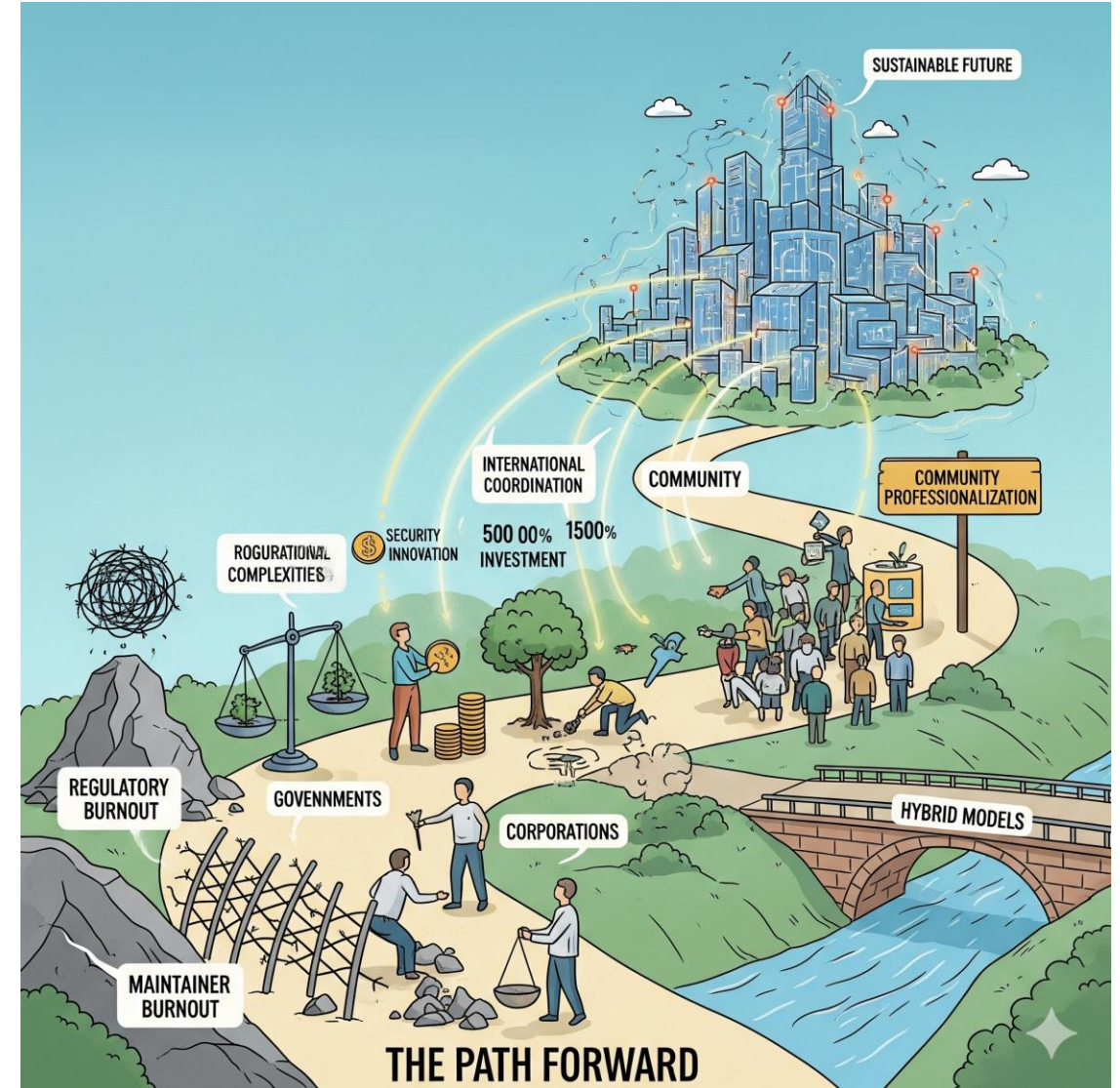
Governments must balance security with innovation; corporations should invest proportionally to derived value.

Community Professionalization

Communities need to professionalize while preserving open principles for sustainable collaboration and growth.

International Coordination and Hybrid Models

International cooperation and hybrid volunteer-corporate-government models are essential to prevent fragmentation and ensure resilience.



Orbis Transportation Network

Goals of our network

Orbis Transportation Network

- **Create an extensive, high-quality, and usable transportation network**

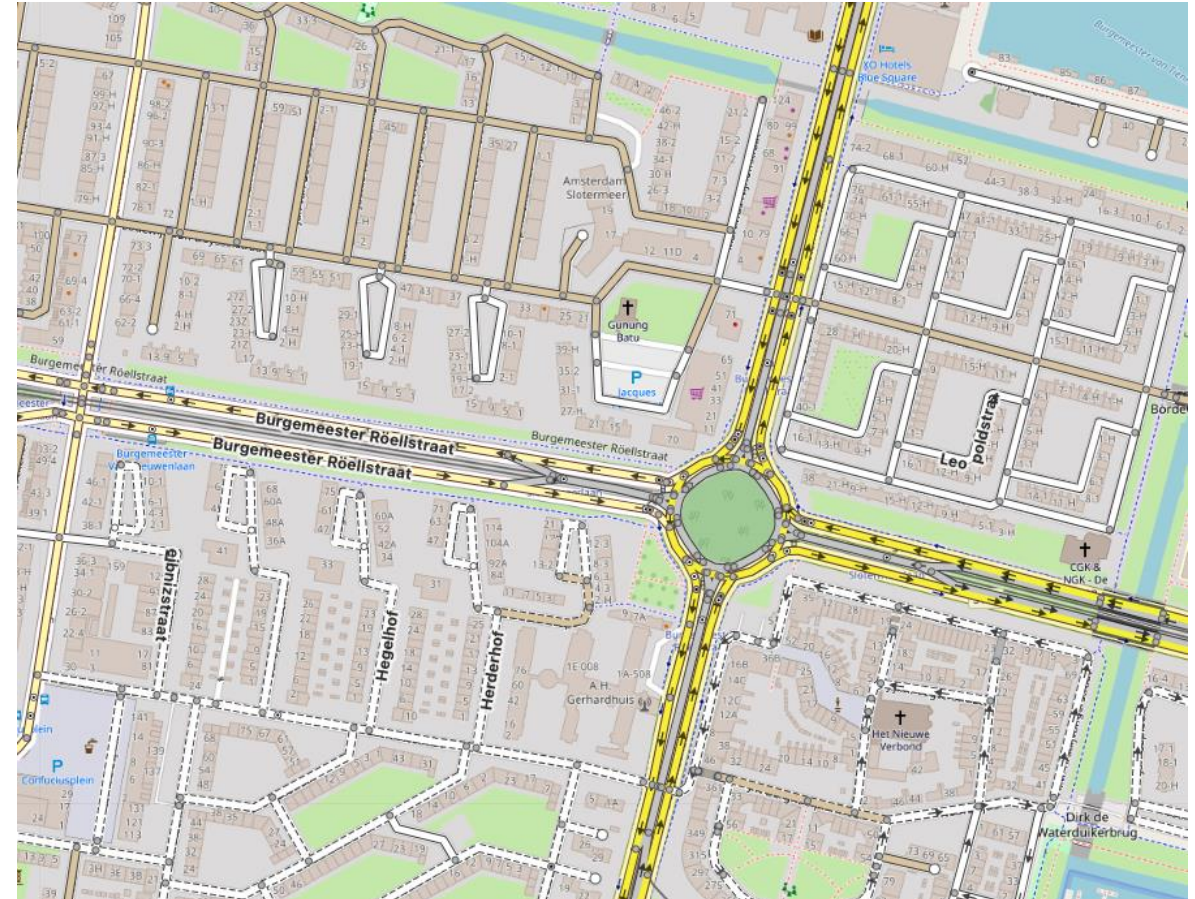
including roads, pedestrian and cycling paths, train, and ferry routes

- **Serve as the backbone of Orbis**

Providing essential structure and functionality

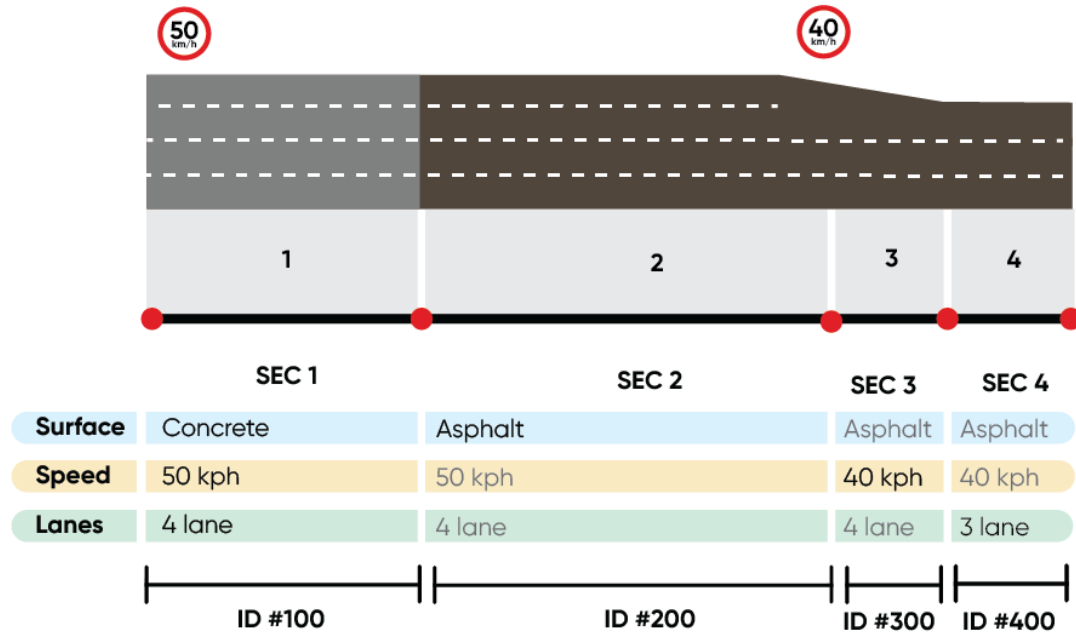
- **Overture Transportation**

TomTom contributes its Transportation Network data (roads, transit, pedestrian routes, etc.) to Overture using stable linear reference systems (Transportation GERS IDs).

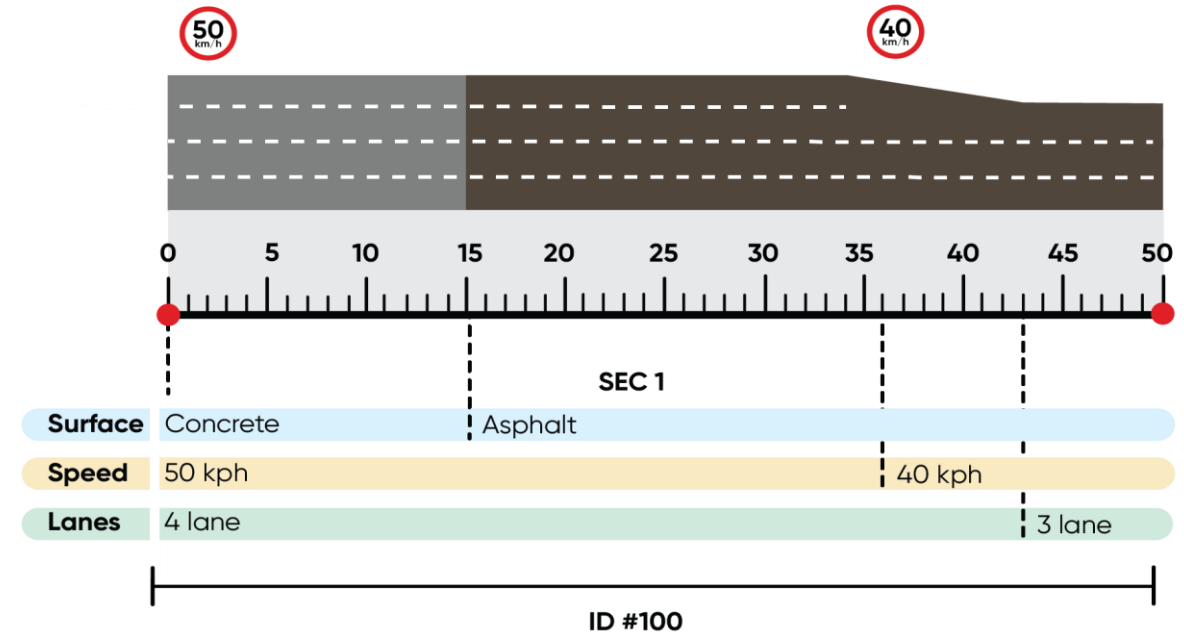


What is Linear Referencing (LR) ?

Orbis Transportation Network



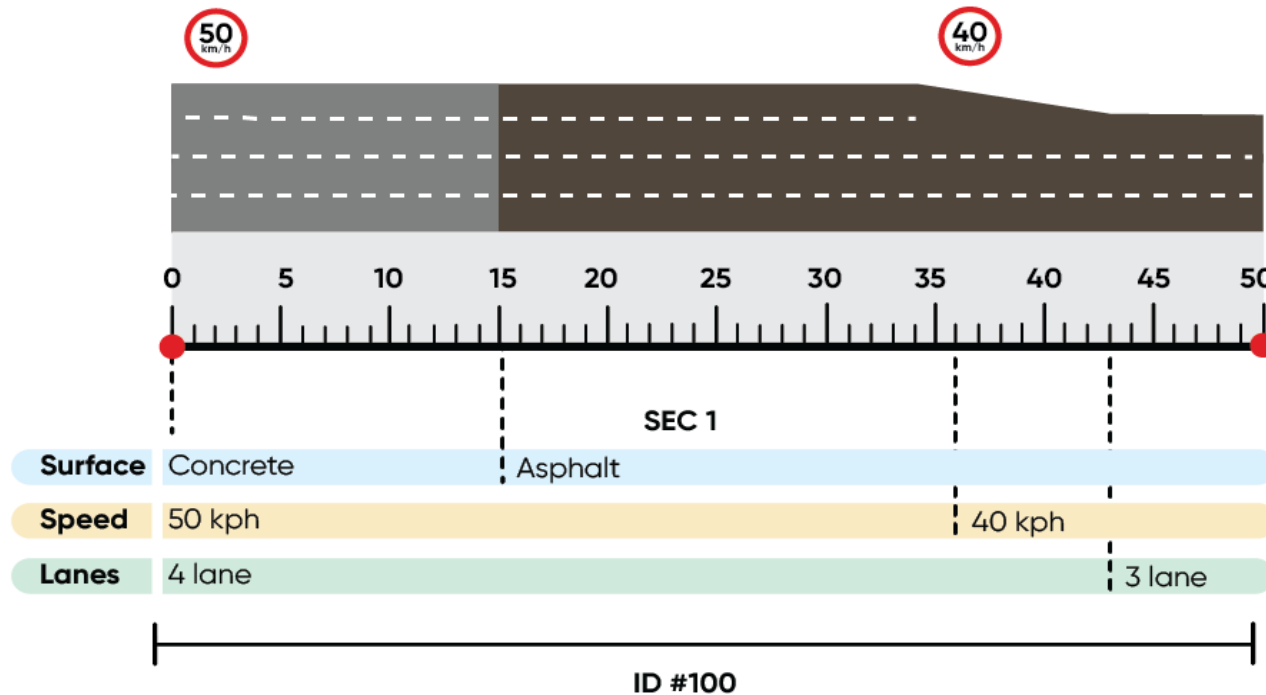
Sectioned Data Model



Linear Referenced Data Model

Why Linear Referencing?

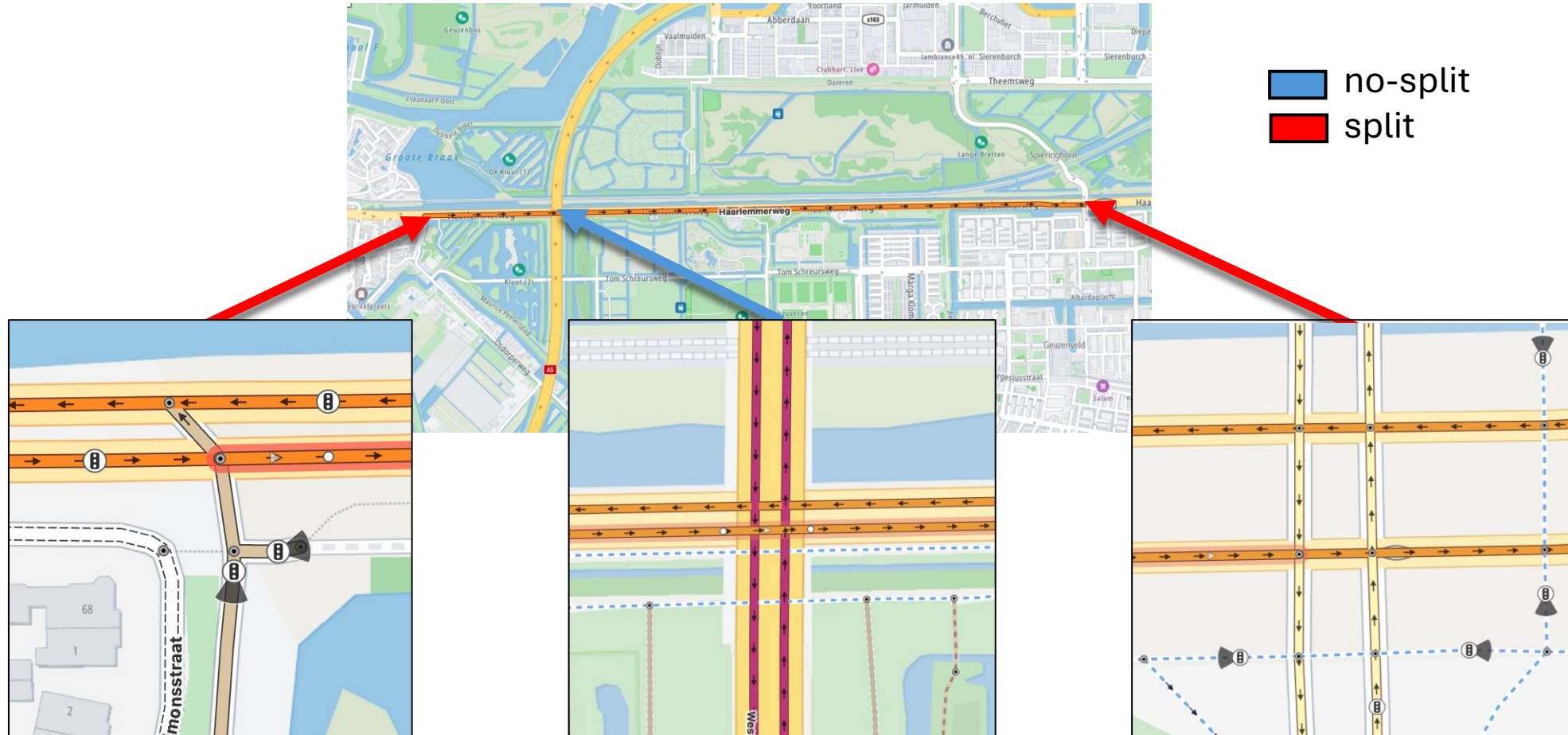
Orbis Transportation Network



- Stable Network IDs
- Optimal Sectioning (Post)

Building a solid base

Orbis Transportation Network



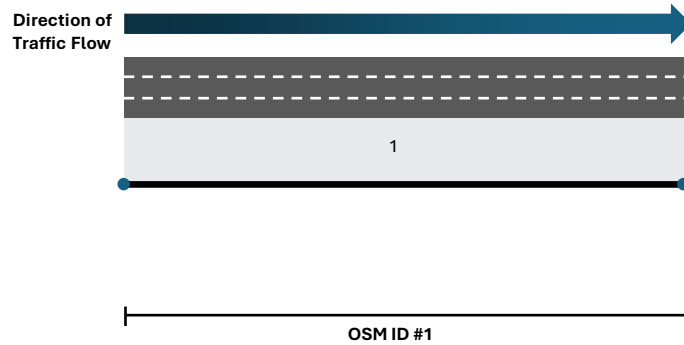
- Consistently introduce **geometry joining and merging** to create a navigable transportation network
- **Assign GERS IDs** to the geometry.

Use case:
Direction of Traffic Flow

Use case: Direction of Traffic Flow

Orbis Transportation Network

Direction of Traffic flow (OSM IDs)

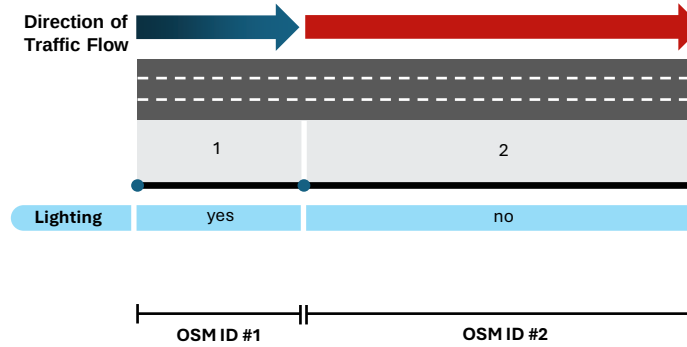
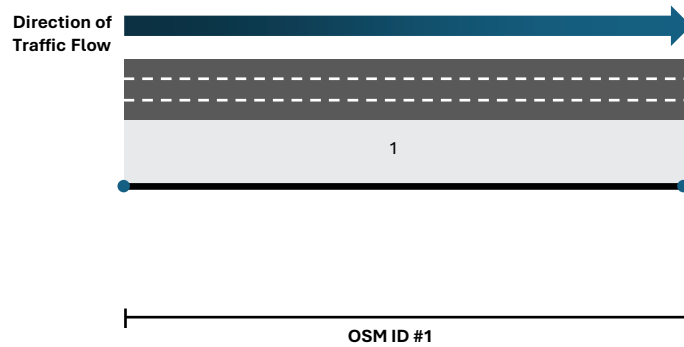


Direction of Traffic flow (GERS IDs)

Use case: Direction of Traffic Flow

Orbis Transportation Network

Direction of Traffic flow (OSM IDs)

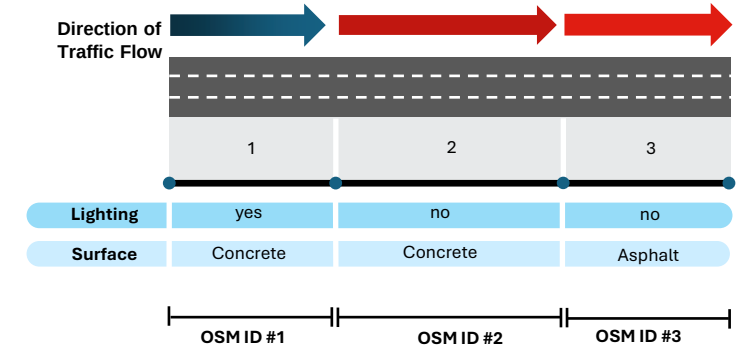
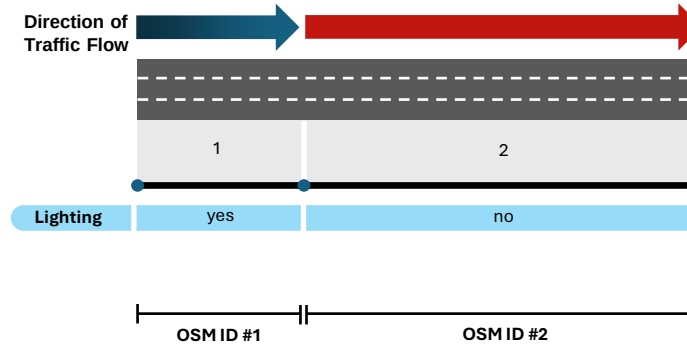
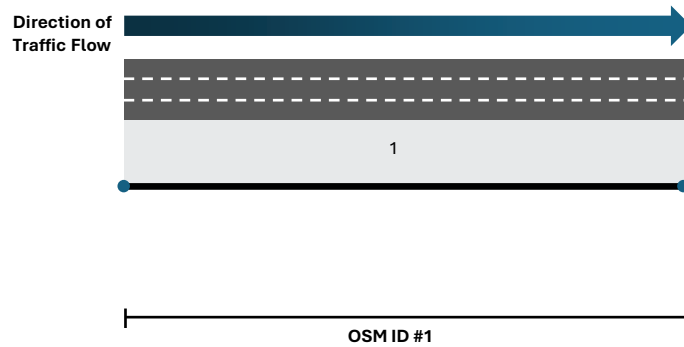


Direction of Traffic flow (GERS IDs)

Use case: Direction of Traffic Flow

Orbis Transportation Network

Direction of Traffic flow (OSM IDs)

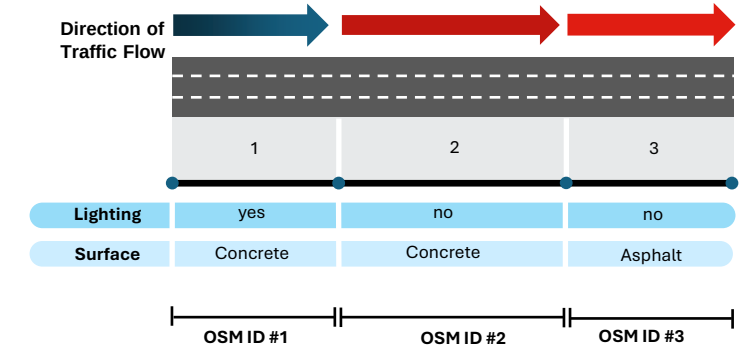
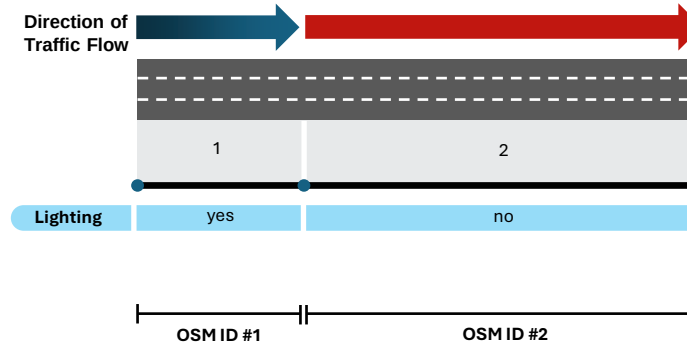
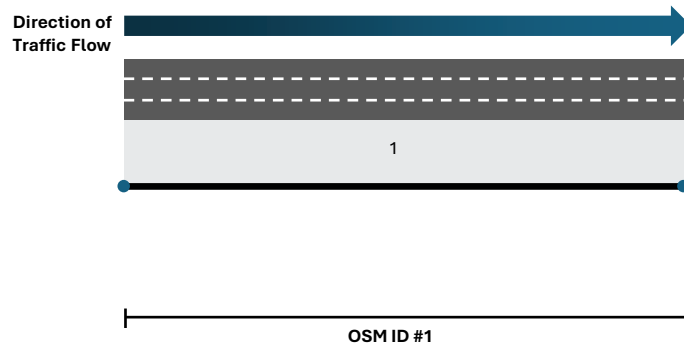


Direction of Traffic flow (GERS IDs)

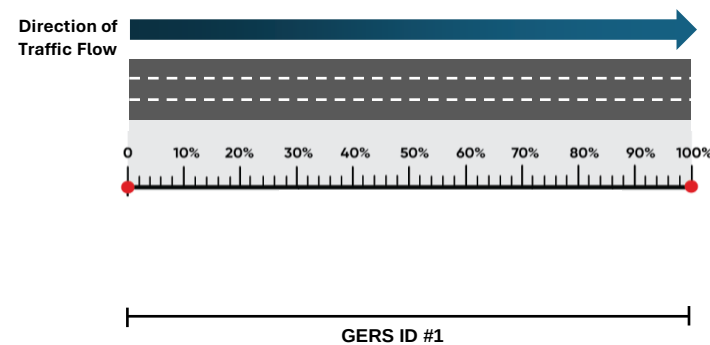
Use case: Direction of Traffic Flow

Orbis Transportation Network

Direction of Traffic flow (OSM IDs)



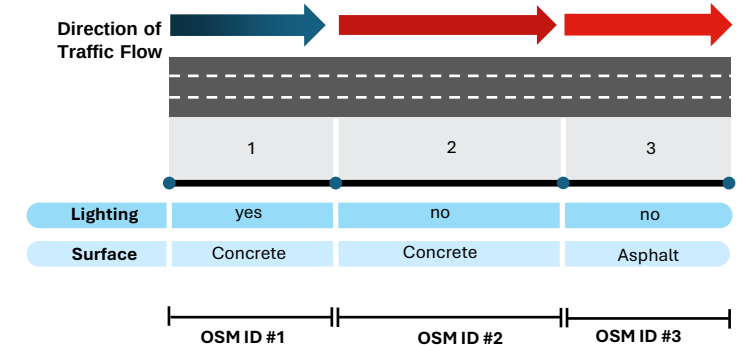
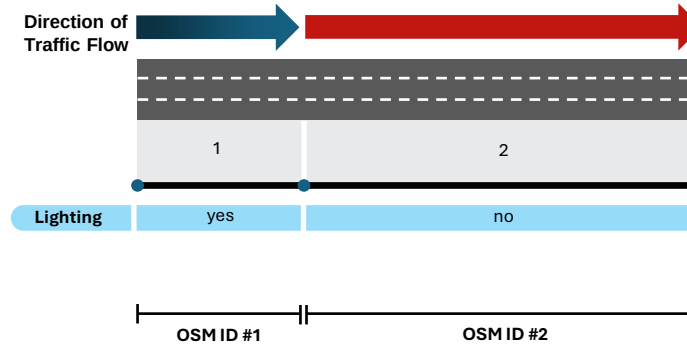
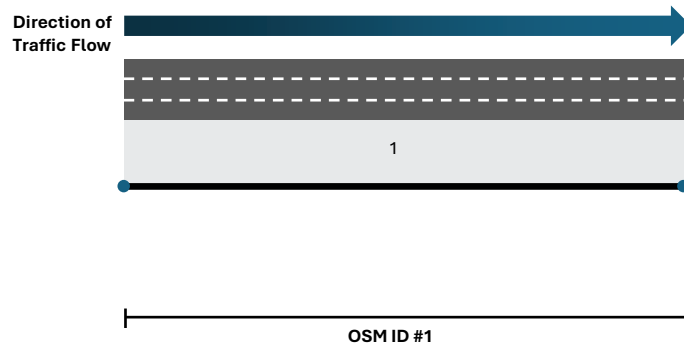
Direction of Traffic flow (GERS IDs)



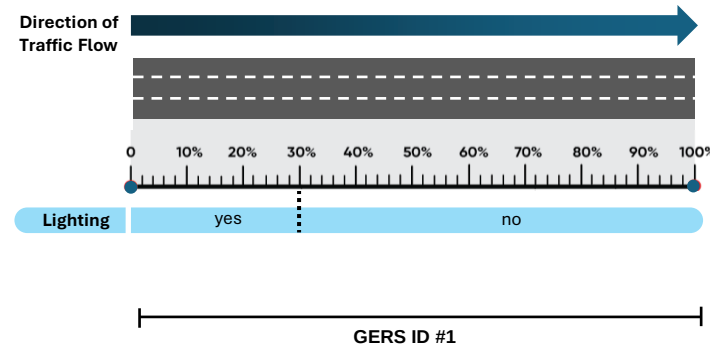
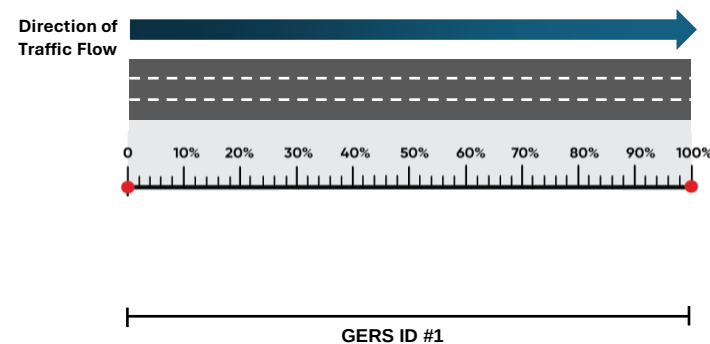
Use case: Direction of Traffic Flow

Orbis Transportation Network

Direction of Traffic flow (OSM IDs)



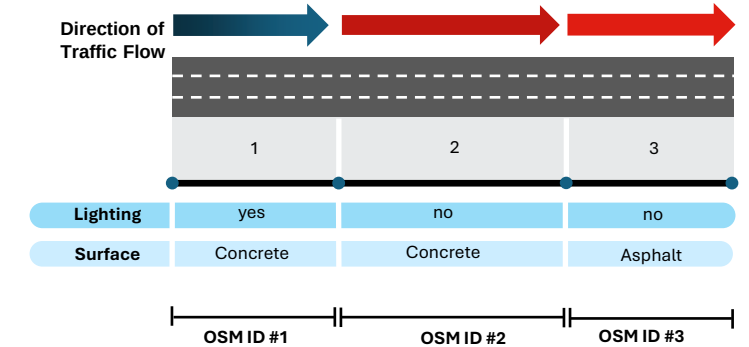
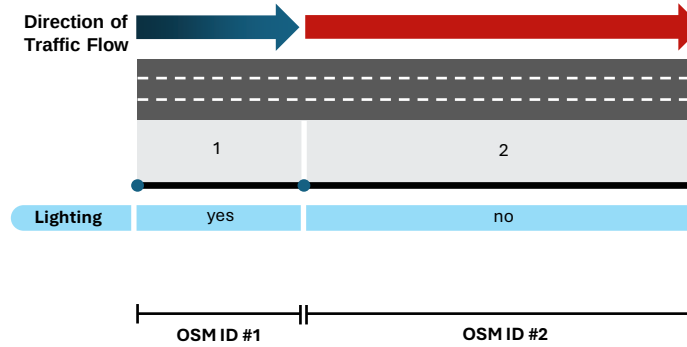
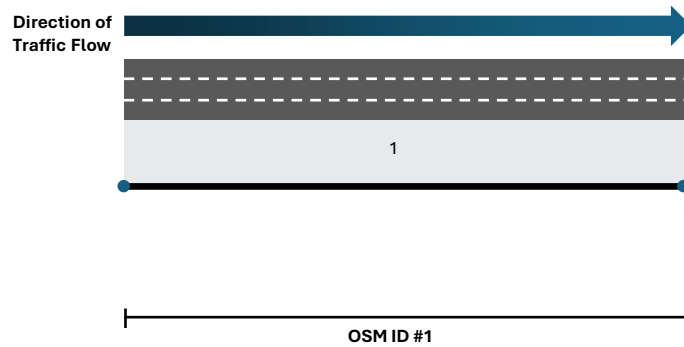
Direction of Traffic flow (GERS IDs)



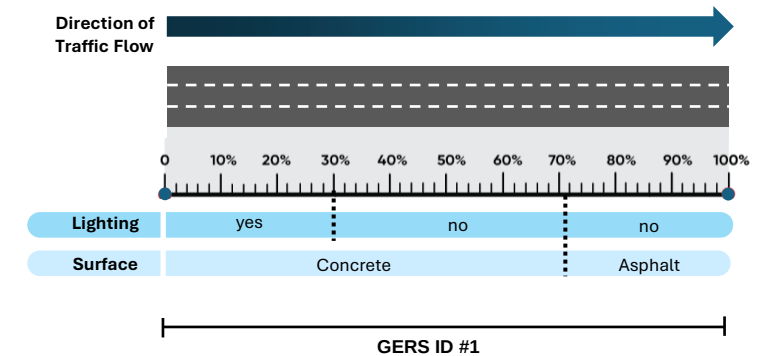
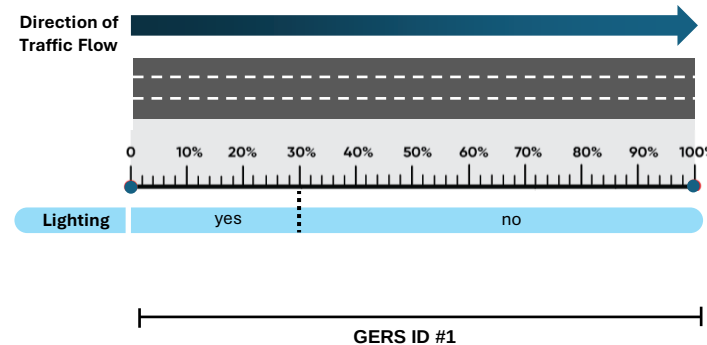
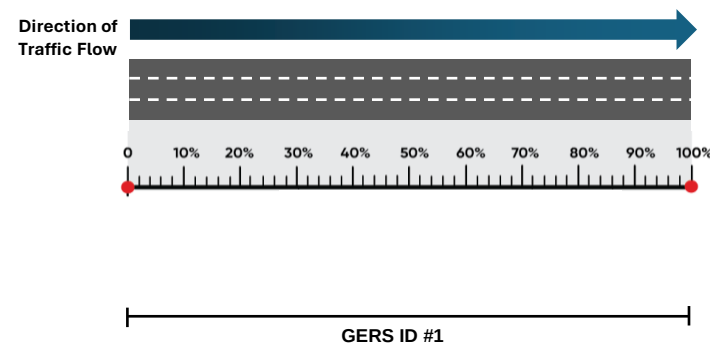
Use case: Direction of Traffic Flow

Orbis Transportation Network

Direction of Traffic flow (OSM IDs)



Direction of Traffic flow (GERS IDs)

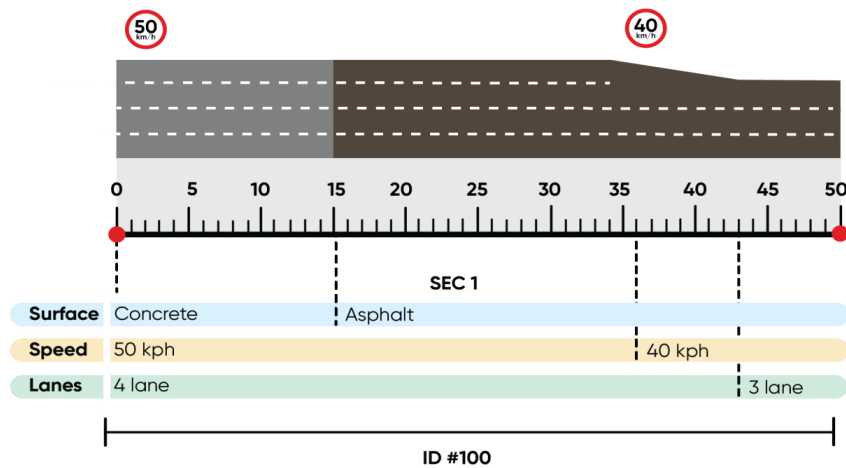


Tooling

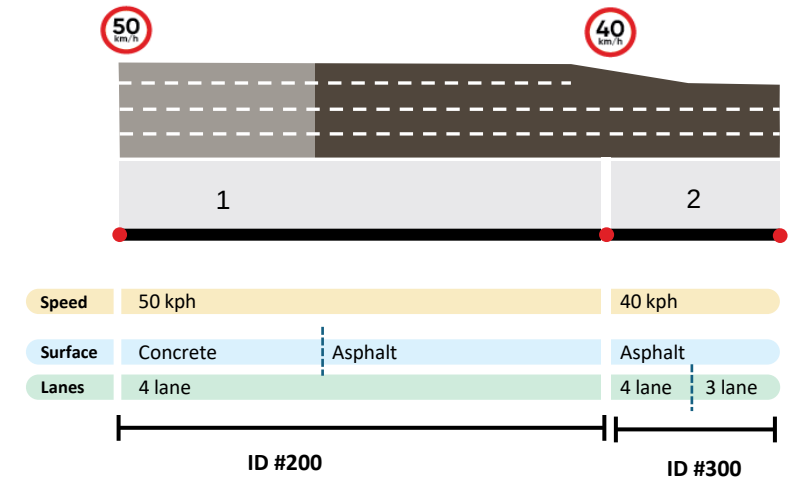
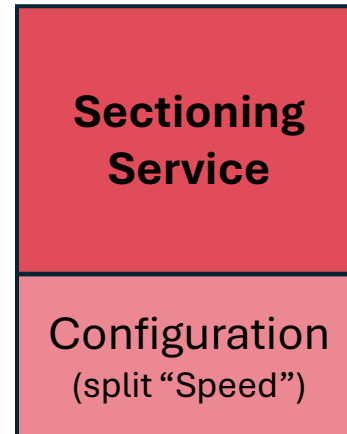
Tools: Sectioning Service

Orbis Transportation Network

Create custom sectioned maps



Orbis OGP - PBF - Overture



Customized map designed for routing that utilizes only speed data.

Tools: Global Entity Matcher (GEM)

Orbis Transportation Network

Connect your geospatial data with Overture's GERS IDs



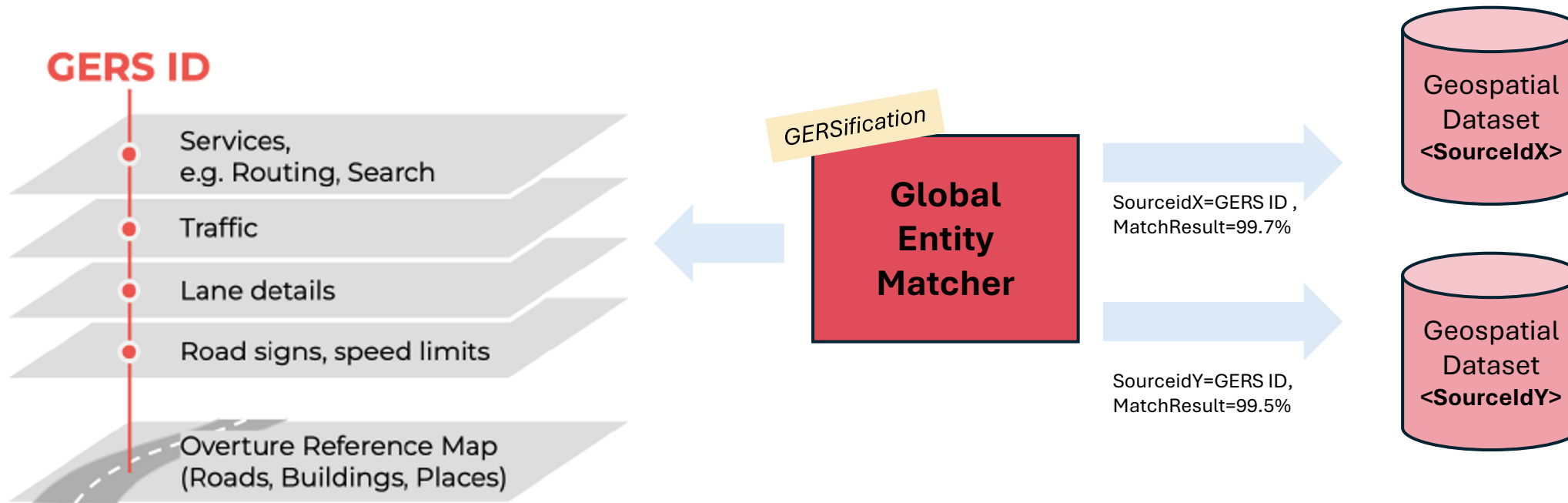
Global Entity Reference System (GERS)

- Universal framework for **structuring & matching** map data across systems
- GERS IDs identify **real world entities** such as road segments
- Simplifies **integrating & exchanging** data layers

Tools: Global Entity Matcher (GEM)

Orbis Transportation Network

Connect your geospatial data with Overture's GERS IDs



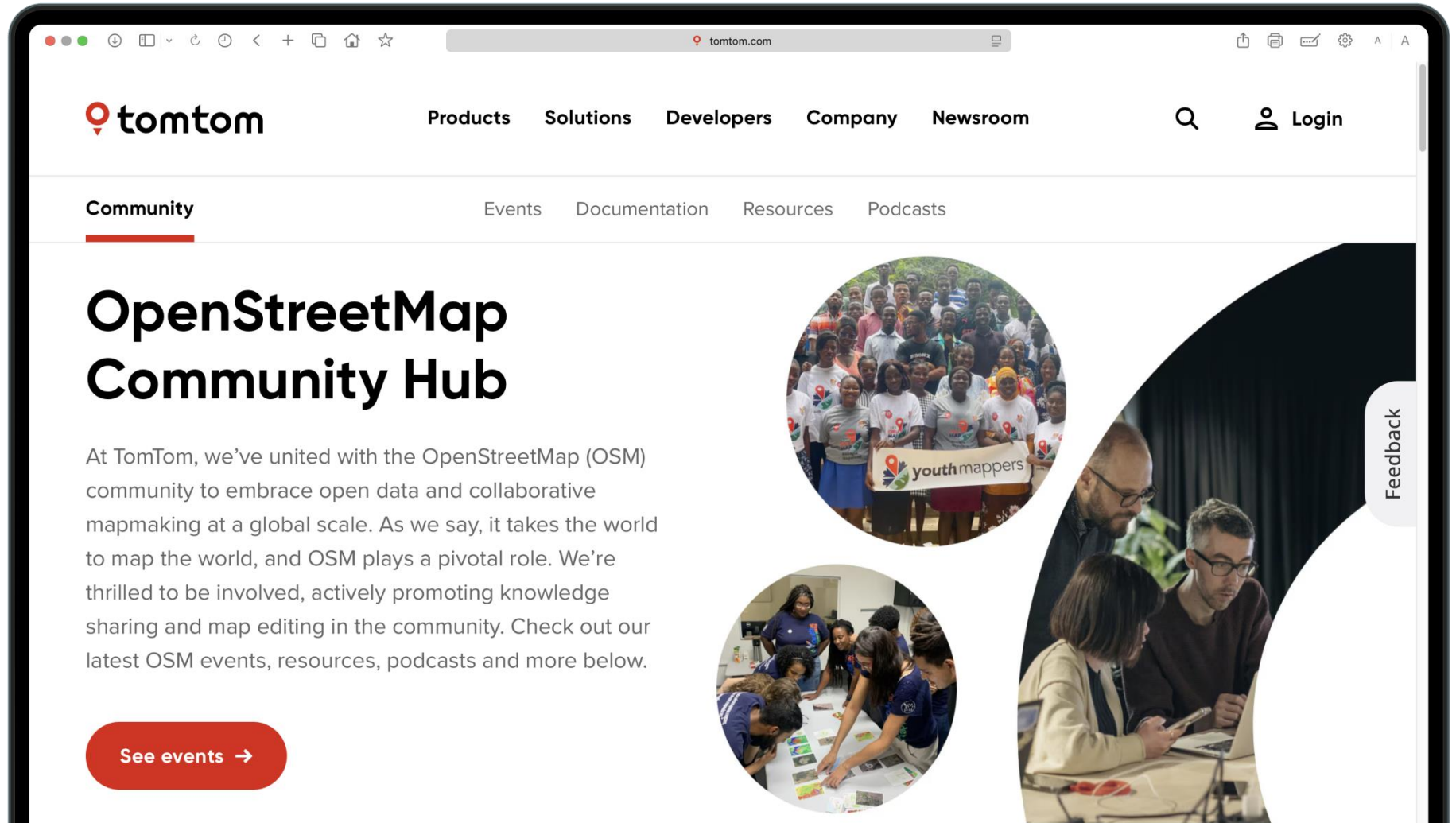
Interested in connecting?



salim.baidoun@tomtom.com



Visit **our webpage:**



Thank you Belgium!

Dutch: Bedankt

French: Merci

German: Danke

